

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 14

Functions of RVs: Technique 2 for CRV

Example 1: Let X_1 and X_2 be *i.i.d.* $U(0, 1)$ random variables. Find the JPDF of $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$.

Example 2: Let X_1 and X_2 be *i.i.d.* $N(0, 1)$ random variables. Find the PDF of $Y_1 = X_1/X_2$.

Moment Generating Function

Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be a RV. The moment generating function (MGF) of $\mathbf{X}$ at $\mathbf{t} = (t_1, t_2, \dots, t_n)$ is defined by

$$M_{\mathbf{X}}(\mathbf{t}) = E\left(\exp\left(\sum_{i=1}^n t_i X_i\right)\right),$$

provided the expectation exists.

Remark: $E(X_1^{r_1} X_2^{r_2} \dots X_n^{r_n}) = \frac{\partial^{r_1+r_2+\dots+r_n}}{\partial t_1^{r_1} \partial t_2^{r_2} \dots \partial t_n^{r_n}} M_{\mathbf{X}}(\mathbf{t}) \Big|_{\mathbf{t}=0}.$

Def: Two RVs $\mathbf{X}$ and $\mathbf{Y}$ are said to have the same distribution, denoted by $\mathbf{X} \stackrel{d}{=} \mathbf{Y}$, if $F_{\mathbf{X}}(\cdot) = F_{\mathbf{Y}}(\cdot)$.

Theorem: Let $\mathbf{X}$ and $\mathbf{Y}$ be two RVs. Let $M_{\mathbf{X}}(\mathbf{t}) = M_{\mathbf{Y}}(\mathbf{t})$ for all $\mathbf{t}$ in a neighborhood around 0, then $\mathbf{X} \stackrel{d}{=} \mathbf{Y}$.

Example 3: Let X_i , $i = 1, 2, \dots, k$ be independent $Bin(n_i, p)$ RVs. Then $\sum X_i \sim Bin(\sum n_i, p)$.

Example 4: Let X_i , $i = 1, 2, \dots, k$ be iid $Exp(\lambda)$ RVs. Then $\sum X_i \sim Gamma(k, \lambda)$.

Example 5: Let X_i , $i = 1, 2, \dots, k$ be independent $N(\mu_i, \sigma_i^2)$ RVs. Then $\sum X_i \sim N(\sum \mu_i, \sum \sigma_i^2)$.

Theorem: X and Y are independent iff $M_{X,Y}(t_1, t_2) = M_X(t_1)M_Y(t_2)$.

Expectation and Variance of a Random Vector

Expectation of a random vector is given by

$$E(\mathbf{X}) = (EX_1, EX_2, \dots, EX_n) = \boldsymbol{\mu}.$$

The variance-covariance matrix of a n -dimensional random vector, denoted by Σ , is defined by

$$\Sigma = [\text{Cov}(X_i, X_j)]_{i,j=1}^n = E(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^t.$$